

QUOTES RECOMMENDATION CHATBOT USING NLP

Epic 2: ENVIRONMENT SETUP

Story 1: Install Rasa and Required Dependencies

1. Introduction

This story focuses on setting up the development environment by installing Rasa and its required dependencies. Rasa is the core framework used to build the Quotes Recommendation Chatbot.

Before installing Rasa, a virtual environment must be created and activated to manage project dependencies separately and avoid conflicts with system packages.

2. Activating the Virtual Environment

After creating a virtual environment, activate it using:

For Windows:

```
venv\Scripts\activate
```

For macOS/Linux:

```
source venv/bin/activate
```

Once activated, the environment is ready for installing required packages.

3. Installing Rasa

Rasa is installed using Python's package manager, pip:

```
pip install rasa
```

This command downloads and installs the latest compatible version of Rasa along with all required dependencies.

4. Components Installed with Rasa

Installing Rasa includes the following major components:

4.1 Rasa NLU

- Handles intent classification
- Performs entity extraction
- Processes user input using NLP techniques

4.2 Rasa Core

- Manages conversation flow
- Controls dialogue management
- Decides the next action based on conversation history

4.3 Supporting Libraries

- Machine learning libraries
- NLP processing tools
- YAML configuration support
- HTTP and REST communication modules

These components work together to build a fully functional conversational AI system.

5. Verifying Installation

To confirm successful installation, run:

```
rasa --version
```

If the version details appear without errors, Rasa has been installed correctly.

6. Conclusion

Successful installation of Rasa and its dependencies ensures that the development environment is fully prepared for building and training the Quotes Recommendation Chatbot.

This step forms the foundation for further stages such as project initialization, intent creation, training data setup, and chatbot development.